Open questions from Grothendieck's Montpellier notes: fourteen statements a formalisation could not reach

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Abstract

The University of Montpellier has put online 178 folders of Alexandre Grothendieck's mathematical working notes. Since August 2026 an edition has transcribed 165 of them with a language model, restated each in current notation, searched the literature for the statements they contain, and formalised what it could in Lean 4 against Mathlib. Most of the statements that look new sit in theories Mathlib does not have yet: étale cohomology, topoi, stacks, abelian schemes, motives. This note collects fourteen of them for mathematicians. Each comes with its folder and page, the context, what the literature search found, the argument the page gives or sketches, the steps the edition added, and what would settle it. A last section takes seven conjectures that are in print under his name, from the standard conjectures to the p-curvature conjecture, and drafts from the pages the route he meant to take to each and where it stops. None of the statements has been checked by a specialist, and several may already be in print; a reference is as welcome as a proof or a counterexample.

1 Why this note

Grothendieck gave his mathematical papers to Jean Malgoire in 1990 and 1995. They were deposited at the University of Montpellier in 2010 and given to it in 2012, and the university placed 178 folders, 16,074 pages by its inventory's count, in open access as facsimiles. They are working notes: drafts in ink and pencil, restarted, struck, annotated, almost never dated. Twenty folders have been edited by hand. The rest had not been transcribed.

Since August 2026 an edition has read the remaining folders with a language model as first transcriber, and written for each folder a *modernised reading*, a restatement in current notation made from the transcription and never from the hand. A register of *findings* lists the statements of the folders that may not stand in the literature, each written as a claim about the literature with what was searched and what would settle it; at the time of writing it holds 456 rows, of which 83 turned out to be in print. Statements were then formalised in Lean 4 against a pinned revision of Mathlib, about two thousand theorems in all. A companion paper describes the formalisation [26].

Lean settles a statement only when the objects it speaks of exist in the library. In October 2026, 203 open findings with no proof were sorted by what Mathlib can reach: 16 were reachable as it stands, 39 in part and 147 not at all, and one was set aside as ill-posed; the 147

are out of reach because they live in étale or crystalline cohomology, topoi, stacks, derivators, motives or Hodge theory [26]. Most of Grothendieck's own work was in those theories. This note hands the most promising of them to people who can prove them with pen and paper, or say where they are already proved.

How to read the entries

Each problem gives its *source*: the folder (a link to Montpellier's facsimile), the pages in the archive's pagination, and the inventory's dating, whose brackets mark an inferred date. « The page » means what Grothendieck wrote, as transcribed; short quotations are his. « The reading » means the edition's modernised reading, made by a language model. « We » marks a step of the edition: an argument, a correction or a search made for this note or for the register, with a language model, and not checked by a specialist unless the entry says so. The entries make no claim of priority. A statement on an undated leaf says when he wrote it down and nothing about whether others had.

The transcriptions themselves are not public: the rights belong to Grothendieck's heirs, and the reading views stay off the site until they authorise them. The facsimiles are public, at Montpellier, and each entry cites the page to look at. Montpellier's servers were down at the end of October 2026; the links follow the archive's own scheme and should work again.

2 Étale cohomology and Hodge theory

Problem 1 (A global bound with a shift of 3)**.** For a field k of characteristic p put $p^{\nu(k)} = [k : k^p]$, and for a scheme Z of characteristic p let $\nu(Z)$ be the supremum of $\nu(k(z))$ over its maximal points. Let X be an affine scheme on which p is not invertible, so that $V(p) \subset X$ has characteristic p. Is it true that

$$\operatorname{cd}_p(X) \;\leq\; \sup\Big(\sup_{x\in X} \operatorname{cd}_p k(x),\; \nu(V(p)) + 3\Big)\,?$$

Source. folder 31, p. 84, *SGA A [SGA 4] (résidus de rédaction)*, inventory dating [1963–1973]. The page is headed « Conjectures sur la cohomologie des schémas affines »; this is its conjecture 2°). The page writes ℓ for the prime; we write p, since it is the characteristic of $V(p)$ and not a prime invertible on X.

Context. M. Artin wrote in SGA 4 X 3.0 that in unequal characteristic one must « certainement tenir compte » of the rank of Ω^1_k, without a bound [46]. The same page states a local form, 4°): for U open in the spectrum of a henselian local domain A of residue characteristic p with fraction field of characteristic 0, $\operatorname{cd}_p(U) \leq \dim A + \nu(k) + \operatorname{cd}_p(k)$. The words around « ouvert » in 4°) are hard to read, so whether U is meant to be affine, and whether it may meet $V(p)$, is not known.

What is known. The local form holds, in a sharper form, for U a non-empty affine open of $\operatorname{Spec} A[1/p]$ with A an excellent henselian local domain and $\operatorname{char} K = 0$: Gabber and Orgogozo prove $\operatorname{cd}_p(U) = \dim A + \dim_p(k)$ [21, Th. 6.1], and Kato's p-dimension satisfies $\dim_p(k) \leq \nu(k) + \operatorname{cd}_p(k)$, which is immediate from their Définition 1.1. Mathew proves that the residue

fields of $\operatorname{Spec} R[1/p]$ have $\operatorname{cd}_p \le d+1$ when R is an excellent normal domain, henselian along an ideal $I \ni p$, with $\dim R_{\mathfrak{p}} + \log_p[\kappa(\mathfrak{p}) : \kappa(\mathfrak{p})^p] \le d$ for $\mathfrak{p} \supseteq I$; the same bound for the affine opens $\operatorname{Spec} R[1/f]$ of $\operatorname{Spec} R[1/p]$ is established in the course of the proof [34, Prop. 4.5]. Globally, SGA 4 gives $\operatorname{cd}_p X \le 2 \dim X + 1$ for X of finite type over $\mathbb{Z}$ [46, X 6.2], and Clausen and Mathew give $\operatorname{cd}_p(X) \le \dim X + \sup_x \operatorname{cd}_p k(x)$ for X qcqs of finite Krull dimension [12, Cor. 3.29]; for $\operatorname{Spec} \mathbb{Z}_p[x_1, \dots, x_n]$, which is not of finite type over $\mathbb{Z}$, the latter gives $2n+3$ where the page's bound gives $n+3$. Gabber and Zavyalov settle the analogous expectation for affinoids [22, Th. 1.7]; the Travaux de Gabber treat ℓ invertible only.

Ideas. The constant cannot be lowered to 2 (our observation): for $X = \operatorname{Spec} \mathbb{Z}$ and p odd, $\nu(V(p)) = 0$ and $\sup_x \operatorname{cd}_p k(x) = \operatorname{cd}_p \mathbb{Q} = 2$, while $\operatorname{cd}_p(\operatorname{Spec} \mathbb{Z}) = 3$ [46, X 6.1]; by Artin–Verdier duality $H^3(\operatorname{Spec} \mathbb{Z}, j_! \mu_p) \cong \mathbb{Z}/p$ [35, II.3.2, 3.3]. A natural route, which is the one of Mathew's proof, is the excision sequence for $V(p) \subset X$ and its complement $X[1/p]$: for X noetherian, $V(p)$ is affine of characteristic p, so of p-cohomological dimension at most 1 [46, X 5.1] (for X not noetherian a limit argument is needed); local cohomology along $V(p)$ can be moved to the henselisation along $V(p)$ by Gabber's affine analogue of proper base change, where Gabber–Orgogozo and Mathew apply. The difficulty is to absorb the dimension terms of those local bounds into $\sup_x \operatorname{cd}_p k(x)$, which the page's bound allows, and to see where the shift 3 comes from: the page does not explain it.

Lean. Not attempted: Mathlib has no étale cohomology.

What would settle it. A printed bound of this shape for an arbitrary affine scheme of mixed characteristic, or a counterexample to the 3.

Problem 2 (Lower bound for a local domain: the exceptions)**.** Let A be a noetherian local domain with fraction field K and residue field k, and r a prime. The page states that

$$\operatorname{cd}_r(K) \;\ge\; \operatorname{cd}_r(k) + \dim A$$

with exactly two possible exceptions: (a) r is the characteristic of K; (b) $r = 2$ and some point x of $\operatorname{Spec} A$ has an orderable residue field $k(x)$ with $\operatorname{cd}_2 k(x)(\sqrt{-1})$ finite. It adds that exception (b) disappears when A is regular, and reduces to « k itself real » when A is catenary and universally japanese. Are these refinements true?

Source. folder 31, p. 22, inventory dating [1963–1973]. The margin gives the example that keeps exception (b) alive: the local ring at the origin of $\mathbb{R}[X,Y]/(X^2+Y^2)$, a real domain whose only real point is the origin (Artin's example after SGA 4 X 2.3). The margin also asks: « cette restriction est-elle essentielle ??? »

What is known. The inequality is SGA 4 X 2.4 when r is invertible in A, with the cruder exception « some residue field orderable » [46]. When r is the residue characteristic and $\operatorname{char} K = 0$, it follows for A excellent henselian from Gabber–Orgogozo §3 [21]. We found nothing on the finer exception (b), on the regular case, on the catenary refinement, or on r the residue characteristic for A neither excellent nor henselian. Serre's *Cohomologie galoisienne* II §4 and Scheiderer's *Real and Étale Cohomology* [45], where cohomological dimension in the presence of orderings is treated, remain to be read.

Ideas. Artin's proof of X 2.4 goes by induction on $\dim A$, down to a henselian local ring of

dimension one through a height-one prime, where X 2.3 and 2.2 apply [46]. The finer exception (b) says that what matters is the virtual cohomological dimension of the real residue fields, which suggests running the same induction with cd_2 replaced by cd_2 of $k(x)(\sqrt{-1})$ at the real points. The page's exception list covers r equal to the residue characteristic in mixed characteristic, which SGA 4 X 2.4 does not.

Lean. Not attempted (Galois cohomology of fields is in Mathlib only in low degree).

What would settle it. A reference for (b) and the two refinements, or an example of a regular local domain where the inequality fails at $r = 2$.

Problem 3 (Hodge types of the deleted neighbourhood of a point)**.** Let Y be a complex algebraic variety of dimension n, $j : X \hookrightarrow Y$ an open immersion with X smooth and dense, $y \in Y \smallsetminus X$, and $H^i = H^i(X \cap B_y) = (R^i j_* \mathbb{Q})_y$ with its mixed Hodge structure (Durfee [15] when $(Y \smallsetminus X) \cap B_y = \{y\}$; in general through Saito's mixed Hodge modules [44]). Then

$$h^{p,q}(H^i) \neq 0 \implies p, q \in [\max(0, i-n+1),\ \min(i, n)],$$

so the weights lie in $[0, 2i]$ for $i \leq n-1$ and in $[2(i-n+1), 2n]$ for $n \leq i \leq 2n-1$ (and $H^i = 0$ for $i \geq 2n$), and the level $\max|p-q|$ is at most $\min(i, 2n-1-i) \leq \min(i, n-1)$. Is this in the literature, and what is the right relative statement for a proper morphism?

Source. folder 12, pp. 115–119, *Généralités (sous-groupes de Galois motiviques)*, inventory dating 1964–[c. 1971]. Page 119 boxes « poids $\leq 2\inf(i,n)$, niveau $\leq \inf(i, n-1)$ » for X regular, $F = \mathbb{Q}_\ell$ and f an immersion with $n = \dim \overline{f(X)}$, in his yoga of weights and levels of ℓ-adic sheaves. The same page asks, with a question mark, for a relative version with $n = d_y(f)$ and level $\leq \min(i, n)$. The typescript says of its own conclusions (p. 96): « la portée des « résultats » de notre travail est nécessairement sujette à caution, puisqu'ils dépendent de conjectures non prouvées à l'heure actuelle » (the Weil conjectures and resolution of singularities).

Context. The page's argument is a Leray spectral sequence for a factorisation of f as an open immersion with normal crossings complement followed by a proper map, conditional on resolution, purity and the Weil conjectures, with tables of weights and levels. The Hodge-theoretic statement above is our translation.

What is known. Deligne bounds the Hodge types of the cohomology of a variety [14, 8.2.4]; Durfee constructs the mixed Hodge structure on $H^i(X \cap B_y)$ and its weight spectral sequence; Durfee–Saito, Steenbrink, Navarro Aznar and El Zein bound weights; for isolated singularities the box is implicit in the literature on links: it follows from semipurity and duality as in [20, Th. 2.6, Def. 2.7]. We did not find the box, or the level bound, for arbitrary y.

Ideas. Our argument: take a resolution $\pi : \tilde{Y} \to Y$, an isomorphism over X, with $D = \pi^{-1}(Y \smallsetminus X)$ simple normal crossings; put $F = \pi^{-1}(y)$ and $F_q = F \cap D^{(q)}$. Then $\dim F_0 \leq n-1$ and $\dim F_q \leq n-q$. The weight spectral sequence $E_1 = H^{k-q}(F_q)(-q) \Rightarrow H^k(X \cap B_y)$ (Durfee's, in the case he treats; we have not checked its form in the general case) and Hodge III 8.2.4 (i), (ii) for the compact F_q put every E_1 term in the box. Locally the lower end moves up by one because $\pi^{-1}(y)$ has dimension at most $n-1$. The cone over a smooth projective V of dimension $n-1$ with $h^{i,0}(V) \neq 0$ attains level i for $i \leq n-1$ and $n-1$ at $i = n$. For

a proper f with fibre dimension d the gain fails, since a smooth family of curves already has H^1 of level $1 = d$; so the relative question of page 119 cannot keep $n - 1$ if $d_y(f)$ is the fibre dimension.

Lean. Not attempted: no Hodge structures on cohomology in Mathlib.

What would settle it. A reference for the box; a relative statement interpolating between the immersion case and the smooth proper case; its ℓ-adic form with level read as divisibility of Frobenius eigenvalues.

Problem 4 (An Ext^2 that vanishes when 2 is invertible: beyond smooth coefficients)**.** On the fppf site of a base on which 2 is invertible, is $\mathcal{E}xt^2_{\mathbb{Z}}(G, M) = 0$ for every finite locally free commutative group scheme G and every quasi-coherent M, not only for the smooth coefficients covered by Breen's theorem below?

Source. folder 129, p. 95, *Notes séminaire Montréal*, inventory dating [1970]. A parenthetical NB after a theorem on Ext over truncated rings: « $\mathrm{Ext}^2_{\mathbb{Z}}(G, M) = 0$ si 2 inv., pour tt G fini loc. libre », with the remark that this simplifies the deformation theory of Barsotti–Tate groups for $p \neq 2$. The word « inv. » is uncertain in the transcription, and the page gives no proof. A year earlier, in a letter to Giraud of 13 June 1969 (IHÉS, fonds Grothendieck–Illusie, FG_IL_2_GIRAUD69, p. 6), he expects the obstruction to lifting a smooth commutative group scheme commutatively over a first-order thickening to lie in $\mathrm{Ext}^2(X_0, \mathrm{Lie}(X_0) \otimes J)$, and asks how to define it; the letter treats smooth groups only and says nothing of a vanishing.

What is known. Breen proved, five years later, that $\mathcal{E}xt^i(G, H) = 0$ for $1 < i < 2p - 2$, for G finite flat and H smooth commutative quasi-projective, when every residue characteristic is at least p and multiplication by every prime $q < p$ is an automorphism of G; and that $\mathcal{E}xt^{2p-2}(\alpha_p, H) \neq 0$ in characteristic p for $H = \mathbb{G}_a, \mathbb{G}_m$ [7]. With $p = 3$ this gives the NB for smooth coefficients, in particular for vector bundles, when 2 is invertible; with $p = 2$ it shows $\mathcal{E}xt^2(\alpha_2, \mathbb{G}_a) \neq 0$, which supports the reading « si 2 inv. ». Illusie places the obstruction to deforming a truncated Barsotti–Tate group over a perfect field in $\mathrm{Ext}^2_{\mathbb{Z}}(G_0, t_{G_0} \otimes J)$ and says that proving directly that it vanishes is out of reach [27, p. 2]; Breen's theorem, which he does not cite, bears on that group.

Ideas. Breen's theorem is stated for the étale sheafification of the Ext sheaves. The remaining cases are a quasi-coherent M that is not locally free, and the passage from Breen's sheaves to the global groups on the fppf site, through the local-to-global spectral sequence.

Lean. Not attempted: no fppf cohomology in Mathlib.

What would settle it. Breen's vanishing for an arbitrary quasi-coherent M, or a counterexample; and whether it gives Illusie's obstruction directly.

3 Algebraic cycles

Problem 5 (Splitting the critical Lefschetz condition)**.** Let X be smooth projective of dimension n over an algebraically closed field, with a Weil cohomology H, a smooth hyperplane section $\varphi : Y \hookrightarrow X$, and L the Lefschetz operator. Write H^{2i} for the cohomology in which codimension-i cycles live, and consider:

- $A'(X)$: for $2i \leq n-1$, a class $x \in H^{2i}(X)$ whose restriction $\varphi^* x$ is algebraic is algebraic;
- $A''(X)$: for $2i \geq n-1$, every algebraic class of $H^{2i+2}(X)$ is $\varphi_* y$ for an algebraic $y \in H^{2i}(Y)$;

and their critical versions A'^0, A''^0, with $2i = n-2$ for n even and $2i = n-1$ for n odd. Let $A^0(X)$ be the critical Lefschetz condition: $L^2 : H^{n-2}(X) \to H^{n+2}(X)$ for n even, $L : H^{n-1}(X) \to H^{n+1}(X)$ for n odd, is a bijection on algebraic classes. In even dimension $A'^0(X)$ and $A''^0(X)$ imply $A^0(X)$ once the critical condition $A^0(Y)$ holds on the section, and in odd dimension the splitting is circular (below, proved in Lean). Is the resulting reduction of the Lefschetz standard conjecture,

$$C(X) \iff C(Y) + A'^0(X \times X) + A''^0(X \times X),$$

correct as stated, that is, can the condition $A^0(Y_T)$ on a hyperplane section Y_T of $T = X \times X$ be dropped?

Source. folder 16, p. 49, *Formalisme algébrique des correspondances et algèbres de Poincaré-Hodge*, inventory dating 1966–1967, in a letter of January 1967. The letters use A, C, D in senses that do not match the later published labels; the conditions above are the reading's restatement. The letter to Coates places the question in the motivic picture (p. 48): « The complete yoga then being that we have the category of motives which is semi-simple ! ».

What is known. The equivalence $C(X) \iff C(Y) + A^0(X \times X)$ and its descent along a chain of hyperplane sections are Kleiman's [30, Th. 2.13]. The splitting of A^0 into the two one-sided conditions was not found in Kleiman 1968 and 1994, Grothendieck 1969, Charles 2013, Ramón Marí, or Milne; André's book [1, ch. 5, §5.2] remains to be read.

Ideas. The page factors the critical operator through the section. For $n = 2m$: $L^2 = \varphi_* \circ L_Y \circ \varphi^* : H^{2m-2}(X) \to H^{2m-2}(Y) \to H^{2m}(Y) \to H^{2m+2}(X)$; for $n = 2m+1$: $L = \varphi_* \varphi^* : H^{2m}(X) \to H^{2m+2}(X)$. What follows is our reconstruction, unchecked by a specialist.

$n = 2m$ even. Let $z \in H^{2m+2}(X)$ be algebraic. By A''^0, $z = \varphi_* y$ with $y \in H^{2m}(Y)$ algebraic. Since $\dim Y = 2m-1$, hard Lefschetz makes $L_Y : H^{2m-2}(Y) \to H^{2m}(Y)$ bijective and weak Lefschetz makes $\varphi^* : H^{2m-2}(X) \to H^{2m-2}(Y)$ bijective, so $y = L_Y \varphi^* x$ for a unique $x \in H^{2m-2}(X)$, and $z = L^2 x$. If $L_Y^{-1} : H^{2m}(Y) \to H^{2m-2}(Y)$ preserves algebraic classes, then $\varphi^* x = L_Y^{-1} y$ is algebraic, and x is algebraic by A'^0. This hypothesis is the critical Lefschetz condition $A^0(Y)$ of the odd-dimensional section. So, for n even,

$$A'^0(X) + A''^0(X) + A^0(Y) \implies A^0(X).$$

$n = 2m+1$ odd. Here $z = \varphi_* y$ with $y \in H^{2m}(Y)$ algebraic, and $x = L^{-1} z$ satisfies $\varphi_*(\varphi^* x - y) = 0$. By the projection formula the kernel of φ_* on $H^{2m}(Y)$ is the orthogonal V of $\varphi^* H^{2m}(X)$, the vanishing cohomology, and $H^{2m}(Y) = \varphi^* H^{2m}(X) \oplus V$. So $\varphi^* x = p(y)$,

where $p = \varphi^* L^{-1} \varphi_*$ is the projection along V, and A'^0 gives $A^0(X)$ as soon as p preserves algebraic classes. But that condition follows from $A^0(X)$ itself. In odd dimension the splitting is therefore a reformulation, unless the decomposition of the algebraic classes of $H^{2m}(Y)$ along $\varphi^* H^{2m}(X) \oplus V$ is supplied otherwise; we do not see that $C(Y)$ supplies it, since $\varphi^* H^{2m}(X)$ is not a Lefschetz piece of Y.

Application to $X \times X$. The displayed equivalence applies the even case to $T = X \times X$, of dimension $2n$. Its hyperplane section Y_T has dimension $2n-1$ and is not $Y \times Y$, and the hypothesis needed is $A^0(Y_T)$, the critical Lefschetz condition in degrees $2n-2$ and $2n$ of a variety of dimension $2n-1$. The direction $\Rightarrow$ holds: $C(X)$ gives $C(X \times X)$ [30], and with Λ_T algebraic a class x with $\varphi^* x$ algebraic is $L_T^{-1} \varphi_* \varphi^* x$, hence algebraic, so $C(X)$ gives $A'^0(T)$, and $A''^0(T)$ likewise. For $\Leftarrow$, an induction on dimension supplies C only below $\dim X = n$, not for Y_T. So what our reconstruction proves is

$$C(Y) + A'^0(X \times X) + A''^0(X \times X) + A^0(Y_T) \implies C(X),$$

and the question is whether the last term can be dropped, or replaced by a condition on varieties of dimension below n. If it cannot, the splitting gains nothing on Kleiman's $C(X) \iff C(Y) + A^0(X \times X)$, where A^0 already bears on $X \times X$ directly.

Lean. The two cases are proved as abstract linear algebra, the cohomology groups being modules and the algebraic classes submodules (`lean/Grothendieck/Folder16.lean`): `split_even` for the display above, `split_odd` and `split_odd_circular` for the odd case. Weak and hard Lefschetz, and the geometry of Y_T, are hypotheses, not theorems.

What would settle it. A reference for the splitting; or a proof that $A^0(Y_T)$ can be dropped from the last display, or replaced by conditions on varieties of dimension below n; or an argument that it cannot, in which case the reduction is circular.

4 Abelian varieties and reductive groups

Problem 6 (Polarised abelian surfaces with even Euler characteristic)**.** Let A/S be an abelian scheme of relative dimension 2 with dual B, and δ a section of $\mathrm{NS}_{A/S}$ with $\chi(\delta)$ even. Is the torsor $\mathrm{Pic}^{\delta}_{A/S}$ under B trivial? Over a field k, this says that the Poonen–Stoll class $c_\delta \in H^1(k, B)$ vanishes [41].

Source. folder 59, pp. 11–13, *Construction des faisceaux inversibles sur schémas de Picard*, inventory dating [années 1960–1970]. Page 11 gives a canonical isomorphism $\mathrm{Pic}^{\delta'}_{B/S} \simeq \tilde{\delta}'_* \mathrm{Pic}^{\delta}_{A/S}$; page 13 composes it with its analogue and concludes, for $n = 2$ and $\chi(\delta)$ even, that Pic^{δ} is trivial. That last line rests on an uncertain « si » and an illegible word.

What is known. Poonen and Stoll define c_λ and prove $2c_\lambda = 0$, with no criterion in terms of $\chi(\lambda)$ or the dimension [41, §4]. We found no such criterion elsewhere.

Ideas. The reading thought the conclusion depends on compatibilities of trivialisations that the folder asks for and never proves. Our argument, unchecked, needs none for mere triviality. Write $c = [\mathrm{Pic}^{\delta}_{A/S}] \in H^1(S, B)$, $\chi = \chi(\delta)$, and $\varphi_L : A \to B$ for the isogeny of a line bundle L of class δ, of degree χ^2.

Step 1, the map. For L of class δ, relatively ample of index 0, the Fourier–Mukai transform $\hat{L}$ is a vector bundle of rank χ on B, and $\varphi_L^*\hat{L} \cong \pi^*(\pi_* L) \otimes L^{-1}$ [37, §3]. Put $N = \det \hat{L}$. Then $\varphi_L^* N \cong L^{-\chi}$ up to a line bundle from S, so $\varphi_L \circ \varphi_N \circ \varphi_L = -\chi\, \varphi_L$, and since φ_L is an isogeny, $\varphi_L \circ \varphi_N = [-\chi]$ on B. The construction commutes with base change, since $\hat{L}$ is computed in degree 0; it defines a morphism of fppf sheaves $\Psi : \mathrm{Pic}^{\delta}_{A/S} \to \mathrm{Pic}^{[N]}_{B/S}$, which needs L only fppf-locally.

Step 2, equivariance. Translating L by $b \in B = \mathrm{Pic}^0_{A/S}$ translates $\hat{L}$: $\widehat{L \otimes P_b} \cong T_b^* \hat{L}$ [37, §3]. Taking determinants, $\Psi(L \otimes P_b) = T_b^* N = N \otimes P_{\varphi_N(b)}$ by the theorem of the square. So Ψ is equivariant along $\varphi_N : B \to A = \mathrm{Pic}^0_{B/S}$.

Step 3, the way back. The same construction on B sends N to $N' = \det \hat{N}$ on A, equivariantly along $\varphi_{N'} : A \to B$. From $\varphi_L \varphi_N = [-\chi]$ and $\deg \varphi_N = \chi^{2g}/\chi^2$ in relative dimension g, one gets $\chi(N) = \pm\chi^{g-1}$ and $\varphi_{N'} = \pm\chi^{g-2}\varphi_L$. This is where the dimension enters: for $g = 2$, N' has class $\pm\delta$, and composing with $L \mapsto L^{-1}$ if the sign is negative, $\Psi' \circ \Psi$ maps $\mathrm{Pic}^{\delta}_{A/S}$ to itself. For $g \neq 2$ the way back lands in the class $\pm\chi^{g-2}\delta$, and the argument does not close.

Step 4, the torsor class. The composite is equivariant along $\varphi_{N'} \circ \varphi_N = \pm\varphi_L\varphi_N = [\mp\chi]$. A morphism from a B-torsor to itself that is equivariant along an endomorphism ψ of B shows $\psi_* c = c$, so $(\chi \pm 1)\, c = 0$. Independently $2c = 0$ [41, §4], through $L \mapsto L \otimes [-1]^* L$. When χ is even, $\chi \pm 1$ is odd and prime to 2, so $c = 0$ whatever the sign.

The points a specialist should check are the sign conventions in Steps 1 and 2, which decide whether the composite is $[\chi]$ or $[-\chi]$ but not the conclusion; the passage from $\varphi_L\varphi_N\varphi_L = -\chi\varphi_L$ to $\varphi_L\varphi_N = [-\chi]$ over a base; and the compatibility of the Fourier–Mukai transform with base change for a polarisation that exists only fppf-locally. Whether the trivialisation obtained is canonical, which is what the folder asks, is a separate question.

Lean. Not attempted: no abelian schemes in Mathlib.

What would settle it. A specialist's check of the argument. One abelian surface over a number field with a polarisation of type $(1, 2)$ and $c_\lambda \neq 0$ refutes the statement. A sharp consequence to test (our heuristic, unchecked): for a genus-2 curve C/k whose Jacobian has real multiplication by $\mathcal{O}_D$, $D \equiv 1 \bmod 8$, defined over k, the theta class is a sum of two classes of even χ, so the statement would force Pic^1_C to have a k-point.

Problem 7 (Kostant's tori in apposition: are the Borels opposite?)**.** Let G be a complex simple adjoint group with Coxeter number h and Weyl group W, and let T, T' be maximal tori in apposition in Kostant's sense [33], with Borel subgroups $B \supset T$ and $B' \supset T'$. Are B and B' always in general position, that is, is $B \cap B'$ a maximal torus? If not for all such quadruples (T, B, T', B'), is there a distinguished conjugacy class of quadruples, stable under the outer automorphisms?

Source. folder 123, pp. 21–22, *[Autour de Kostant]*, letter to Kostant of 22 October 1969: « is it true that for some, or for all, such quadruples, B and B' are « in general position » […] ? Is there in any sense a *distinguished* conjugacy class of such quadruples […] ? » The letter states without proof that the quadruples form $|W|^2/(h\, z\, \varphi(h))$ conjugacy classes, z the order of the centre of the simply connected cover, and (in a footnote in his hand) that $N(T) \cap N(T')$ is an extension of $(\mathbb{Z}/h)^*$ by $\mathbb{Z}/h \times Z(\tilde{G})$.

What is known. The count, which this note listed as open until 26 October 2026, follows from standard theory (our argument, unchecked by a specialist). Let $x = \rho^\vee(e^{2\pi i/h})$, the principal element of T, and T' the centraliser of a cyclic element $E' = \sum_i c_i e_{\alpha_i} + c_0 e_{-\theta}$. An element $t \in T$ normalising T' sends E' into $\mathfrak{h}' \cap \mathfrak{g}^{\mathrm{Ad}(x)=\zeta}$, which is the line $\mathbb{C}E'$, so $\alpha_i(t) = \theta(t)^{-1} = \lambda$ with $\lambda^h = 1$: $T \cap N(T') = \langle x \rangle$, of order h. The image of $N(T) \cap N(T')$ in W is the set of w with $w(x) \in \langle x \rangle$: the stabiliser of x is the group $\Omega \cong P^\vee/Q^\vee$ of order z (in the adjoint group x has a disconnected centraliser), x is conjugate to x^k exactly for k prime to h [33, 48], and each such w lifts into $N(T')$, because regular semisimple elements of the eigenspaces $\mathfrak{g}^{\zeta^j}$, j prime to h, have the same centralisers as those of $\mathfrak{g}^\zeta$ and their Cartan subspaces are conjugate under $Z_G(x)^\circ = T$. Hence $|N(T) \cap N(T')| = h\, z\, \varphi(h)$; the stabiliser of a quadruple is $T \cap T' = 1$, and the count follows. A computer count over $\mathbf{F}_q$ (`computations/kostant`) agrees: 2, 4 and 12 orbits for A_2, B_2 and G_2. For general position we found nothing.

Ideas. For A_2 all 36 pairs (B, B') are opposite: the eigenvectors of E' are the Fourier vectors $(1, \omega^k, \omega^{2k})$, and the standard flag is transverse to every flag they span. In general the question is the Bruhat cell of (B, B') for B' running over the $|W|$ Borels through T': whether it is always the open cell.

Lean. Only the arithmetic: the announced number $\mathrm{card}(W)^2/(h\, z\, \varphi(h))$ is an integer for every simple type of rank at most 8 and for E_6, E_7, E_8, F_4, G_2, and equals 1, 2, 4, 12 for A_1, A_2, B_2, G_2 (`lean/Grothendieck/Folder123Kostant.lean`). The geometry, and the relative position of B and B', are not attempted; the computer count above can be extended to that position in B_2 and G_2, which we have not done.

What would settle it. The relative position of B and B' for all quadruples, by an argument or by computation in each type.

5 Topoi and stacks

Problem 8 (Two classifying topoi that are not coherent)**.** The theories of commutative rings with irreducible spectrum (nilradical prime) and of rings with zero-dimensional spectrum are geometric. The page says their classifying topoi, subtopoi of the coherent classifying topos of rings, are not coherent, their presenting topologies on affine schemes of finite type over $\mathbb{Z}$ not being quasi-compact. Is that right, and do these topoi have enough points?

Source. folder 161-2, pp. 39, 41–42, *Algèbre universelle*, inventory dating [c. 1963–1973]. Page 39: « Cette topologie n'est pas quasi-compacte »; page 42, margin: « topologie non cohérente ! A-t-elle assez de pts ?? »; page 41, margin: « encore un topos non cohérent (bien que sous-topos d'un topos cohérent) ».

What is known. Locally coherent toposes have enough points (Deligne, [46, VI, 9.0]); these theories are geometric but, as presented, use the infinite disjunction hidden in « nilpotent ». We found neither non-coherence nor the existence of points stated for them, in Johnstone [28] or in Caramello's criteria for De Morgan classifying toposes, which take quotients of the theory of rings as examples [10, §5].

Ideas. Non-coherence needs more than the presentation: one must show that no coherent axiomatisation exists, for instance by exhibiting a cover in the classifying topos with no finite subcover. For points, Barr's theorem gives a Boolean cover, not points; a model-theoretic argument, or an explicit set of local rings detecting the theory, is needed.

Lean. Not attempted: classifying topoi are not in Mathlib.

What would settle it. A proof or reference for non-coherence, and an answer to the question about points.

Problem 9 (Foliations and their quotient topoi)**.** Is there an equivalence between the category of pairs (X, R), with X a topological space and R a local equivalence relation on X, locally open with locally connected fibres, and the category of triples $(X, \mathcal{Y}, f)$, with $f : \mathrm{Top}(X) \to \mathcal{Y}$ a locally open geometric morphism with locally connected fibres to a « multiplicité topologique », a 2-category whose Hom-categories are discrete?

Source. folder 119, pp. 144–145, dated « Mars 1986 » on p. 144. Under « Th »: « On a $C \rightleftarrows C'$ », both arrows marked $\approx$. No proof. A third condition on f is partly illegible; the reading suggests it may be a simple connectedness of the fibres.

What is known. SGA 4 IV, Exercise 9.8.3 j)–k), gives the functor $(X, R) \mapsto (\mathrm{Top}(X) \to \mathrm{Top}(X/R))$ [46]; Rosenthal states that the topos $\mathrm{sh}(X; r)$ exists for strictly open r (Th. 3.9, proof in his thesis) and proves it is an étendue for strictly regular r [43]; Kock and Moerdijk show that every étendue is of the form $\mathrm{sh}(M, r)$, construct the inverse $a \mapsto \mathrm{Lker}(a)$ and show the round trips are identities under simple connectedness or kernel-pair conditions [31], and study monodromy versus holonomy [32]. We found no statement of an equivalence of categories.

Ideas. The two functors are there; the problem is the right conditions on both sides so that they are mutually inverse, and which quotient topos the page means: the topos of R-invariant sheaves (sheaves on a monodromy groupoid) or the holonomy quotient, which Kock and Moerdijk distinguish.

Lean. Not attempted: no geometric morphisms of topoi in Mathlib.

What would settle it. A correct statement with proof, or Moerdijk's Zaragoza lectures [36], which we could not read and where it might be.

Problem 10 (Realising the symmetry invariant of a Picard stack)**.** Let M, N be abelian sheaves on a topos, and ${}_2N$ the 2-torsion of N. Is every homomorphism $M \to {}_2N$ the symmetry invariant of a Picard stack pinned by M and N?

Source. folder 134-1, p. 5, letter to Deligne, Lodève, 27 August 1974 (Breen's copy, forwarded by Deligne): « Je sais prouver (sauf erreur) que tout hom. $M \to {}_2N$ provient d'un champ de Picard convenable (épinglé par M, N) (a priori l'obstruction est dans $\mathrm{Ext}^3(X; M, N)$, mais un argument « universel » prouve qu'elle est nulle) ». No argument is written. On p. 6 the letter asks his own version of this problem: « Question : Le triangle exact (T) et la suite exacte (*) sont-ils connus par les compétences (Quillen, Breen, Illusie…) ? Connaissent-ils

des variantes « supérieures » ? » (are they known to the experts, and do they know higher variants?). Grothendieck withdrew the vanishing in a letter to Breen of 5 February 1975 (IHÉS, fonds Grothendieck–Illusie, 1ILL21, p. 1; the copy of this letter in the appendix to *Pursuing Stacks*, folder 134-2, p. 37, skips its beginning). He grants that Breen's scepticism about the exact sequence of the Deligne letter was founded, and accepts Breen's reading of the invariant as an Ext^2 of a functor of M (written $X(M)$) into N. He keeps an obstruction homomorphism $\mathrm{Hom}(M, {}_2N) \to \mathrm{Ext}^3(M, N)$, found by a geometric route, and the reduction to the universal case, $M = N$ the universal sheaf of $\mathbb{F}_2$-vector spaces, which he calls correct; but the vanishing of $\mathrm{Ext}^i(M, N)$ for $i \geq 1$ in that case came, he writes, from a « calcul complètement canulé ».

What is known. Strict Picard stacks are classified by Ext^2 (Deligne, [46, XVIII, 1.4]); Breen classifies Gr-stacks by H^3 and gives the universal coefficient spectral sequence $\mathrm{Ext}^p(H_q(B), A) \Rightarrow H^{p+q}(B, A)$ [8, §6]. No printed source we found states the realisation, or refutes it on a topos.

Ideas. Two sketches disagree. The reading realises the invariant by hand for M free, on the split stack $M \times BN$ with an antisymmetric form b such that $b(e_i, e_i) = q(e_i)$, and passes to general M through a resolution. We doubt that step, which we have not checked: the construction depends on a basis. On the classifying topos of abelian groups, a Picard stack pinned by the generic group U and $U/2U$, with the projection as invariant, would make Moore spectra truncated above π_1 functorial in finitely presented abelian groups; but the stable Moore-space category on finitely generated groups is a non-split linear extension, split on cyclic groups [3, Rem. 1.5.5]. If that translation is right, the obstruction does not vanish universally. The letter of 1975 agrees with the second sketch on one point: the « universal » argument of 1974 rested on a miscalculation. It does not say whether the obstruction is non-zero. The statement does hold over a point and for M constant and free.

Lean. Not attempted: no stacks on a topos in Mathlib.

What would settle it. The obstruction $\mathrm{Hom}(M, {}_2N) \to \mathrm{Ext}^3(M, N)$ computed in a universal case: the identity of the universal sheaf of $\mathbb{F}_2$-vector spaces, to which the letter of 1975 reduces, or the differential $\mathrm{Hom}(M/2, N) \to \mathrm{Ext}^3(M, N)$ of the universal coefficient spectral sequence on the classifying topos of abelian groups. Since the 1974 argument is withdrawn, a non-zero value there would answer the problem in the negative.

Problem 11 (Morphisms of ramification domains are fully faithful)**.** Over a scheme S, a « domaine de ramification » is a stack of multigaloisian categories over the étale site of S with a continuous geometric realisation into relative schemes, subject to axioms R1 and R2 (pp. 190–193). For a domain (R, r) defined by a scheme Z with a finite group G of operators, is $\mathrm{Hom}((R, r), (R', r'))$ equivalent to the category of pairs (P', α), P' a galoisian object of R'_S of group G and α a G-isomorphism $r'(P') \simeq Z$, so that every homomorphism of domains is fully faithful?

Source. folder 29, p. 197, *Groupe fondamental [autour de SGA 4 et SGA 7]*, inventory dating 1959–1969: Proposition 6, stated without proof, with the corollary « nécessairement pleinement fidèle ». The axiom R2 of pp. 190–193 is itself not verified by the edition. Earlier in the folder (p. 24) he notes where removing the auxiliary choices leads: « il semble qu'on soit

infailliblement conduit à prendre comme catégorie [...] la catégorie [...] elle-même qu'on se propose de construire et d'étudier ! », comparing it with sites and their topoi.

What is known. Nothing found. Tonini's work on stacks of ramified Galois covers [49, 50] is where an equivalent statement would sit; neither surfaced in our search, which is not the same as their not containing it.

Ideas. The corollary says that a ramification domain maps into another only as a subdomain. The proof should reduce to the case (R, r) given by (Z, G), where a morphism is determined by the image of the universal G-torsor.

Lean. Not attempted: no étale site or stacks of categories in Mathlib.

What would settle it. A proof of Proposition 6 from R1–R2, or a reference.

6 Moduli of curves and Galois actions

Problem 12 (The Galois image, seen on one curve)**.** Let X be an anabelian curve over a field K of finite type over $\mathbb{Q}$, and $\hat{\Pi}_1(X, I)$ its profinite fundamental groupoid on a finite set I of base points. The page lists necessary conditions for a vertex-fixing automorphism Θ of $\hat{\Pi}_1(X, I)$ to come from $\mathrm{Gal}(\bar{K}/K)$: it lifts to $\hat{\Pi}_1(X', I')$ for every finite étale cover X' of X defined over K; it preserves the kernels of $\hat{\pi}_1(X', x') \to \hat{\pi}_1(Y, f(x'))$ for every K-morphism $f : X' \to Y$; it is compatible with morphisms between covers; and it satisfies the loop condition at the cusps with a single multiplier. Are these conditions sufficient?

Source. folder 142, pp. 94–95, 98–101, *Action de Galois sur Teichmüller*, inventory dating [from 1978]. Page 95 conjectures the conditions sufficient « lorsque X est (disons) une courbe algébrique anabélienne, avec K extension de type fini de $\mathbb{Q}$ »; page 99 calls the loop condition « sûrement la plus importante de toutes, et de très loin ! ». Pages 98–99 are largely illegible.

What is known. Ihara's question and the Oda–Matsumoto conjecture characterise the Galois group as the automorphisms of the geometric fundamental group functor on all varieties compatible with all morphisms; Pop proved it in that form, and finite tripod variants, built from varieties of dimension greater than one with finitely many dominant rational maps to $\mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$ [42, Def. 2.2, Th. 2.9]. Hoshi–Mochizuki and others give injectivity for a single hyperbolic curve; Collas proposes a combinatorial model GT^{NF} as a candidate for $G_{\mathbb{Q}}$, left to forthcoming work [13, §4.3.3]. We found no characterisation attached to one curve and its covers, with base points and the inertia condition.

Ideas. Pop's variants need a base of dimension greater than one, so they do not cover the covers of one curve; that is what the page's loop condition at the cusps would have to supply. Here « anabelian curve » is the page's word for a hyperbolic curve.

Lean. Not attempted: no étale fundamental group in Mathlib.

What would settle it. A proof of sufficiency, a counterexample, or a published variant that implies it.

Problem 13 (The deleted neighbourhood of a normal crossings divisor)**.** Let X be a smooth complex analytic space (or orbifold) with a normal crossings divisor Θ, and $V^*_{\Theta,X}$ the deleted tubular neighbourhood of Θ. Is $V^*_{\Theta,X}$ the homotopy colimit of the semi-simplicial system of deleted tubular neighbourhoods of the strata of the unfolding of Θ, so that its fundamental groupoid is the amalgamated sum $\tilde{\Pi}_1 \leftarrow \tilde{\Pi}_{2,1} \rightarrow \tilde{\Pi}_2$ of the groupoids of the torus bundles over the open strata of codimension 1 and 2? And hence the same for $\Pi_1(X \smallsetminus \Theta)$ whenever $X \smallsetminus \Theta$ and its end have equivalent fundamental groupoids?

Source. folder 147, pp. 148–151, *Structure à l'infini des* $M_{g,\nu}$. The folder is undated; the inventory dates its group (141–149) [from 1978]–1983. We date it to the winter of 1982–1983, because its formalism is the one Grothendieck describes to Illusie in a letter of 22–25 January 1983 (IHÉS, fonds Grothendieck–Illusie, 1ILL35, p. 27) and in a *rapport d'activité* of which he sent Illusie a copy on 29 March 1983 (1ILL41, p. 4; copy in folder 119, pp. 141–142). Page 149 states a « Th. de recollement » as an equivalence of topoi and corrects it at once: « en fait, c'est une équivalence d'homotopie ». Pages 150–151 give the corollaries, with a purity argument for neglecting the strata of codimension 3. No proof is written.

Context. The folder describes the boundary of $\overline{M}_{g,\nu}$ by stable graphs, lists the strata of modular dimension at most 2, and writes on its p. 29 (archive p. 59) that a more or less complete understanding of the discrete π_1 in these small cases « donnera la clef pour une compréhension complète dans le cas général »: the two-level principle of the Teichmüller tower, about a year before the *Esquisse d'un programme*. Applied to $X \smallsetminus \Theta = M_{g,\nu}$, the corollaries would express the mapping class group through the strata of codimension 1 and 2; that application is the reading's, not the page's.

What is known. The two-level principle for the discrete tower was proved by Hatcher, Lochak and Schneps [23]. For the general gluing statement, the places to look are the real oriented blow-up and the Kato–Nakayama space of the log structure, whose boundary is a torus-bundle model of the deleted neighbourhood, and Looijenga's and Boggi's work on the boundary of $\overline{M}_{g,n}$. We did not find the statement in this form.

Ideas. The purity step reads: removing a subset of real codimension at least 3 does not change π_1. For $M_{g,\nu}$, the hypothesis of the last corollary (end and interior with the same Π_1) should follow from the simple connectivity of the curve complex (Harer).

Lean. Not attempted: no homotopy colimits of spaces or stratified spaces in Mathlib.

What would settle it. A reference for the gluing theorem, or a proof, and the check of the hypothesis for $M_{g,\nu}$.

7 Pseudoline arrangements

A pseudoline arrangement in the real projective plane is *bigonal along* one of its pseudolines D when D carries exactly two vertices of the arrangement, so that every other pseudoline meets D at one of them. In the page's notation, II_4 is four pseudolines in general position, II_5 five in general position, $\mathrm{II}'_{3,2}$ a pencil of three and a pencil of two without the line joining their centres, and III_6 the complete quadrangle (the six lines joining four points, no three collinear); Figure 1 draws all four.

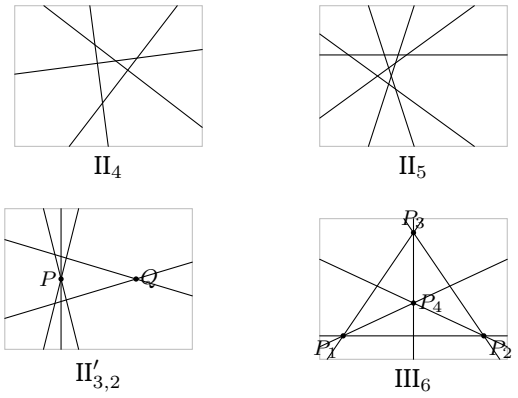


Figure 1: The four arrangements of the dichotomy, drawn with straight lines in an affine chart chosen so that every crossing is visible. In $\mathrm{II}'_{3,2}$ the line PQ is absent; in the complete quadrangle III_6 the six lines join $P_1, \dots, P_4$, and the triangle $P_1P_2P_3$ has every other line through one of its vertices, the case that needs the second step of the answer to Problem 14 (Figure 2).

An arrangement is *bigon-free* when none of its pseudolines is bigonal, and four pseudolines are in *general position* when no three of them pass through one point. The page's boxed dichotomy, which this note listed as an open problem until 26 October 2026, has a short proof, and so does the numbering lemma of page 28; we record it because the problem below depends on it.

Proposition (Grothendieck, folder 80, pp. 20, 24, 28; the proof is ours)**.** *A bigon-free arrangement $\mathcal{A}$ either is II_4 or contains II_5, $\mathrm{II}'_{3,2}$ or III_6; and its pseudolines can be numbered, starting from a II_4 or a III_6, so that every partial arrangement is bigon-free.*

Proof. Four pseudolines in general position. $\mathcal{A}$ is not a near-pencil, since each pencil line of a near-pencil carries only the centre and its point on the transversal; so it has a simple crossing $p = a \cap b$ [29, Th. 3.6], see also [5, Th. 1.3]. Since a is not bigonal it carries vertices $q_1, \dots, q_k$ besides p, with $k \geq 2$. Take $c \neq a$ through q_i and $d \neq a$ through q_j, $i \neq j$. Then $\{a, b, c, d\}$ is in general position as soon as $c \cap b \neq d \cap b$: indeed c, d miss p, meet a at different points, and $c \cap d$ lies neither on a nor on b. If this fails for every choice, all these pseudolines meet b at one point r; two pseudolines meet only once, so each q_i lies on a and on a single other pseudoline c_i, all through r, and every pseudoline meets a at p or at some q_i. Then $\mathcal{A} = \{a, b, c_1, \dots, c_k\}$ and b carries only p and r, so it is bigonal, which is excluded.

The dichotomy. Let $Q \subset \mathcal{A}$ be four pseudolines in general position, and $\ell \in \mathcal{A} \smallsetminus Q$. If ℓ passes through none of the six vertices of Q, then $Q \cup \{\ell\}$ is a II_5; if through exactly one, a $\mathrm{II}'_{3,2}$. Otherwise ℓ passes through exactly two, which must be an opposite pair $\{x \cap y, z \cap w\}$ of the quadrilateral $Q = \{x, y, z, w\}$ (two vertices on a common side would force ℓ to be that side). Call such an ℓ a diagonal of Q; there are at most three. If $\mathcal{A}$ contains neither II_5 nor $\mathrm{II}'_{3,2}$, then $\mathcal{A}$ is Q plus some diagonals. One diagonal alone would be bigonal; two diagonals together with Q form a III_6 (the four triple points are the vertices of the quadrangle and the six pseudolines its sides); with three, $\mathcal{A}$ contains a III_6 again.

The numbering. Adding pseudolines never removes a vertex from a pseudoline already there, so it suffices that each pseudoline, when added, meets the earlier ones in at least three points. A pseudoline meets the six sides of a III_6 in at least three points: its vertices have

multiplicity at most three, and two triple points share a side, so two vertices cover at most five sides. So if $\mathcal{A} \supseteq \mathrm{III}_6$, start there and add the rest in any order. Otherwise $\mathcal{A}$ contains at most one diagonal δ of Q: start from Q, add every pseudoline other than δ (each meets Q in at least three points), and add δ last, when it has all its vertices in $\mathcal{A}$, at least three. □

This agrees with a computer enumeration of all arrangements of at most nine pseudolines (`computations/bigon`). The dichotomy is also checked in Lean for combinatorial maps (generalised maps of Euler characteristic 1 whose flags are labelled by the pseudolines; `lean/Grothendieck/Folder80Maps2.lean`, `map_dichotomy`); the numbering and the passage from a topological arrangement to its map are not formalised. We have not found the statement in print; it is elementary, and probably folklore among people who work on arrangements.

Problem 14 (A bigon-free arrangement is determined by its skeleton: the three-pencil case)**.** Is a combinatorial arrangement of pseudolines in the real projective plane without bigonal pseudolines determined by its 1-skeleton together with the decomposition of the edges into pseudolines, functorially: is the functor from bigon-free arrangements to their decomposed skeleta fully faithful, so that every automorphism of the decomposed skeleton comes from the arrangement? The Lean proof in combinatorial maps covers every arrangement in which each triangle has a fourth pseudoline through none of its vertices (see below). What it leaves is a bigon-free arrangement with a triangle $P_1P_2P_3$ whose three vertices are multiple and through which every other pseudoline passes, so that the arrangement lies in the three pencils at P_1, P_2, P_3; the complete quadrangle III_6 is the first example.

Source. folder 80, pp. 39–54. The statement of page 54 (« En résumé ») is clean. It composes a full faithfulness theorem (p. 39) for arrangements with an orientation of every non-concurrent triple (pp. 41–49, conditions (A1)–(A4)) with a canonical reconstruction of these orientations from the skeleton (pp. 49–54, (A5), (A6), with the complete quadrangle as residual case). Conditions (A1)–(A6) are said to hold « bien entendu » and not proved.

What is known. Line arrangements in the projective plane are determined up to isomorphism by their circular sequences up to local equivalence (Goodman–Pollack), and the graph of a simple projective arrangement of at least four pseudolines is 4-connected [19, Th. 5.2.2, 5.5.1]. Embeddings of graphs in the projective plane are unique at face-width at least 3 and are classified at face-width 2 and 3 [51, 38, 39]. For simple arrangements the question reduces to the plane, by our argument, unchecked by a specialist: deleting one labelled pseudoline leaves a simple Euclidean arrangement; its graph, with one vertex added and joined to the vertices of degree less than four, is 3-connected and planar [16], so by Whitney's theorem every isomorphism of such graphs (which preserves degrees) comes from a homeomorphism of the sphere, and the pseudolines are the paths through opposite edges [17, Lemma 1]. Lean now proves more (see below): multiple points are allowed as long as every triangle has a fourth pseudoline avoiding its vertices. We found no statement for arrangements with multiple points in print. The question, with the three-pencil case singled out, is also asked on MathOverflow (question 516410, `mathoverflow.net/q/516410`).

An answer. Yes, for every bigon-free arrangement, by the following argument, which no specialist has checked. It goes through oriented matroids. Let φ be an isomorphism of decom-

posed skeleta from $\mathcal{A}$ to $\mathcal{A}'$, and σ the bijection it induces on the pseudolines. Since every pseudoline carries at least three vertices, φ is determined by σ: a vertex goes to the vertex on the images of its pseudolines, and an edge is fixed by its two ends and its label. Choose chirotopes χ and χ' of the two arrangements. Along L_i, φ keeps the cyclic order of the vertices up to reversal, so by the representation theorem in rank 2 the contractions at i and at $\sigma(i)$ differ by a sign s_i and a reorientation ε^i:

$$\chi'(\sigma i, \sigma j, \sigma l) = s_i \, \varepsilon^i_j \, \varepsilon^i_l \, \chi(i,j,l)$$

for every three pseudolines i, j, l without a common point. Put

$$\psi(i,j,l) = \chi'(\sigma i, \sigma j, \sigma l)\, \chi(i,j,l)\, s_i s_j s_l.$$

It is symmetric, and at each i it has the form $\psi(i,j,l) = \delta^i_j \delta^i_l$, with $\delta^i_j = \varepsilon^i_j s_j$. We show that $\psi = 1$.

Suppose first that a fourth pseudoline m avoids the vertices of the triangle i, j, l. The rule at i gives $\psi(ijl)\,\psi(ijm)\,\psi(ilm) = 1$, so the product Π of ψ over the four triples of $\{i,j,l,m\}$ is $\psi(jlm)$. The same holds at j, l and m, so all four values equal Π, and $\Pi = \Pi^4 = 1$. This is condition (A5) in another form.

Otherwise every other pseudoline passes through a vertex of the triangle. Call its sides a, b, c and its vertices $P_1 = b \cap c$, $P_2 = a \cap c$, $P_3 = a \cap b$. The pseudoline a carries a third vertex, and a pseudoline through it other than a can only pass through P_1; let x be one. The rule at a gives $\psi(abc) = \psi(abx)\,\psi(axc)$. Because b carries a third vertex, some pseudoline $y \neq a, c$ passes through P_2, and it avoids the vertices P_3, P_1 and $a \cap x$ of the triangle a, b, x. In the same way a pseudoline $z \neq a, b$ through P_3 avoids the vertices of the triangle a, x, c. By the first case both factors are 1 (Figure 2). In the complete quadrangle, for the triangle with sides P_1P_2, P_2P_3, P_3P_1, the three pseudolines x, y, z are the sides through P_4.

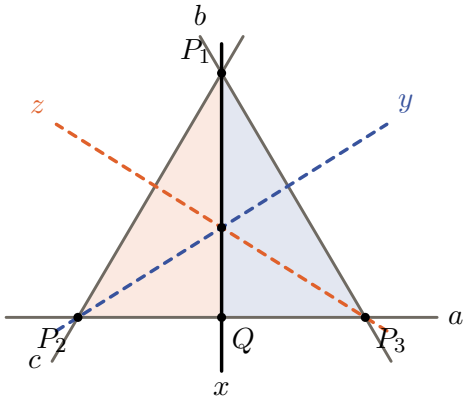


Figure 2: The second step of the answer to Problem 14, on the complete quadrangle. The triangle a, b, c has no fourth line avoiding its vertices. The line x through P_1 splits it into the blue triangle a, b, x, which the line y through P_2 avoids at its vertices, and the orange triangle a, x, c, which the line z through P_3 avoids. So $\psi(abc) = \psi(abx)\,\psi(axc) = 1$.

Hence $\chi' \circ \sigma$ is the reorientation of χ by s, and the Folkman–Lawrence representation theorem gives a homeomorphism of the projective plane that carries L_i to $L'_{\sigma(i)}$ for every i [18], [19, §5.2]. This homeomorphism induces σ, and therefore φ. For arrangements of lines, Good-

man and Pollack had shown that two arrangements are isomorphic when they have circular sequences in which each line meets the others in the same or the opposite order [19, Th. 5.2.2]. The argument above uses the same data for pseudolines and gives full faithfulness. The face-width bound with a uniqueness theorem for embeddings would not have given the statement: the graph of III_6 is K_7 minus a triangle and has six inequivalent polyhedral embeddings in the projective plane (`computations/bigon`).

Lean. Proved, multiple points included, whenever every triangle has a fourth pseudoline through none of its vertices (`full_of_fourth'`), in the combinatorial maps of page 60 with its axiom that each pseudoline goes straight through each vertex (`lean/Grothendieck/Folder80Maps3.lean`). The steps: the functor is faithful (`faithful`); a skeleton isomorphism lifts exactly when it preserves the face paths of length three (`lift_iff`, page 41); the orientation of a triangle is read on the skeleton when a fourth pseudoline avoids its vertices ((A5), `triangle_iff`); the side on which an edge leaves another is read by bounding triangles ((A4), `side_tri`); and two edges at a vertex of any multiplicity are adjacent iff no edge lies between them ((A1)–(A3), `between_iff`, `SkelIso.corner_map`). A triangle with a vertex on exactly two pseudolines always has a fourth one (`fourth_at`), which gives the statement when every triangle has such a vertex (`full_of_simpleCorner`), in particular for simple arrangements (`full_simple`). In that model, (A6) is not formalised, nor is the passage from a topological arrangement to its map. The sign argument above, from the rule along each pseudoline to the reorientation, is proved in `lean/Grothendieck/Folder80Signs.lean` (`eq_zero_of_fourth`, `eq_zero`, `reorientation_of_locallyEquivalent`), for any incidence structure in which two lines meet in one point, every point lies on two lines and every line carries three points. So is the rank-2 step (`const_add_cut`, `reorientation_of_sameSeparation`): on a complete multipartite graph with at least three parts, a sign function whose sum over every 4-cycle vanishes is a constant plus a cut, and along a pseudoline these sums are the separation relation of the vertices. What is left outside Lean is the Folkman–Lawrence theorem, and the reading of the separation relation on an actual arrangement.

A computer test (`computations/bigon/three_pencil.py`) finds no counterexample in the open case up to nine pseudolines: for the 132 bigon-free arrangements of at most nine pseudolines that lie in the three pencils at the vertices of a triangle, every automorphism of the labelled skeleton comes from the arrangement, and no two of them have isomorphic labelled skeleta. The sign argument depends only on which pseudolines are concurrent. Its linear algebra over $\mathbb{F}_2$ (`computations/bigon/obstruction.py`) leaves no room for anything other than a reorientation on any of the $37\,213$ bigon-free intersection lattices of at most nine pseudolines that the enumeration finds.

What would settle it. A check of the argument above by a specialist, in particular of the form in which it uses the two representation theorems: the contraction of a simple oriented matroid of rank 3 at an element is read as the cyclic order of the vertices along the corresponding pseudoline.

8 The large conjectures, and how the notes approach them

The fourteen problems above may not be in the literature. Seven conjectures of the notes are, under Grothendieck's name, and the companion paper lists where each first appears in the fonds [26]. This section reads the same pages for how he meant to prove them. For each

conjecture it gives the statement in today's form, the pages, a draft of the route the pages take, where the route stops, and what is known now. The drafts are the edition's, made from the transcriptions with a language model. They report what the pages do, and say so when a step comes from the modernised reading instead. None of the seven is solved here, and no route is complete on the page.

Conjecture 1 (The standard conjectures)**.** Let X be smooth projective of dimension n over an algebraically closed field, with a Weil cohomology H and the Lefschetz operator L of an ample class. *Lefschetz type*: the inverse of $L^{n-i} : H^i(X) \to H^{2n-i}(X)$ is induced by an algebraic correspondence. *Hodge type*: on the primitive algebraic classes of $H^{2i}(X)$, $2i \leq n$, the form $(-1)^i\langle L^{n-2i}x, x\rangle$ is positive definite.

Source. folder 14, pp. 43–46, letter to Serre of 27 August 1965, printed in their correspondence; folder 14, pp. 105–113; folder 16, pp. 32–54, with letters to Coates (pp. 43–49), inventory dating 1966–1967; folder 12, pp. 158–166; folder 15, p. 169.

The route. The 1965 letter fixes the order of the argument. A Lefschetz condition on cycles gives a Künneth formula for cycles, hence characteristic polynomials of algebraic endomorphisms with integer coefficients, hence the Weil conjectures except the absolute values of the eigenvalues; the positivity gives the absolute values, and the two together make the motives semisimple. He says he can derive the full Lefschetz condition from a weak, restriction-type one, provided it holds over base fields that are not algebraically closed. Folder 16 then sets up an induction along hyperplane sections $\varphi : Y \hookrightarrow X$. The identity $\Lambda_Y = \varphi^*\Lambda_X^2\varphi_*$ (p. 39) gives $C(X) \Rightarrow C(Y)$, with C the Lefschetz-type conjecture; conversely, «Moyennant $C(Y)$, $C(X)$ équivaut à la condition» that $\Lambda_X\varphi_*$ be algebraic (p. 41), which is the Lefschetz condition on $X \times Y$ in the critical degree only. Iterating down a chain of sections reduces C for all varieties to that critical condition for all varieties (pp. 44–45); Problem 5 is the next step, which splits the critical condition into two one-sided ones (p. 49). A letter to Coates gives a second reduction (pp. 47–48): $C(X)$ follows from integral characteristic polynomials and the existence of *some* algebraic isomorphism $H^{2n-i} \to H^i$, by Cayley–Hamilton. For positivity the route goes through algebras: the rings $\mathrm{End}(h^i(X))$ should carry a positive involution, hence be semisimple, hence the motives form a semisimple abelian category (folder 12, p. 158; folder 16, p. 48); and one line proposes «Positivité à la Hodge en se ramenant à des questions de diviseurs» by induction (folder 15, p. 169).

Where it stops. The letter of 1965 misses a way of moving a cycle to infinity in the affine complement $X \smallsetminus Y$, which the weak-to-full step needs. The letter to Coates says of the induction: «Thus the main problems arise already for 1-cycles on threefolds» (folder 16, p. 46). The 1965 conditions are stated modulo algebraic equivalence, and in that form they are false (Griffiths), as the modernised reading notes. For positivity there is only the one line of folder 15.

Today. Open in general. The Lefschetz type is known for curves, surfaces and abelian varieties [30], and the Hodge type in characteristic 0 by the Hodge–Riemann relations. Over $\mathbb{C}$, Arapura proves the Lefschetz type for every variety motivated by curves or surfaces in André's sense, among them uniruled threefolds and unirational fourfolds [2, Cor. 4.3], and for Hilbert schemes of points on a surface and moduli spaces of stable bundles on a curve, under the conditions stated there [2, Cor. 7.2, 7.5]. Beilinson shows that in characteristic 0 the standard

conjectures follow from the existence of a motivic t-structure [4, §1.4–1.5], and that over $\mathbb{C}$ they are equivalent to a weak form of Suslin's conjecture on Lawson homology [4, §2.2, Prop.]. One step of his proof, over $\mathbb{C}$ with rational coefficients, is the reduction of the letter to Coates: if for each $i > 0$ some correspondence c induces an isomorphism $H^{n+i} \to H^{n-i}$, the Lefschetz type holds, the inverse of L^i being $f(A)\,c$ for $A = c \circ L^i$ and a polynomial f with $f(A)A = 1$ on H^{n-i} [4, §2.3, proof of (b)]. Deligne proved the Weil conjectures without them. An OpenAI preprint of 6 October 2026, not peer-reviewed and not formalised, announces the Hodge conjecture for complex abelian varieties with complex multiplication and deduces the Hodge type for abelian varieties in every characteristic [40].

Conjecture 2 (The generalised Hodge conjecture, in its ℓ-adic form)**.** For X smooth projective, the coniveau filtration $\mathrm{Filt}^j H^i(\bar{X}, \mathbb{Q}_\ell)$, made of the classes that vanish on the complement of a closed subset of codimension $\geq j$, is the realisation of the largest sub-motive of $h^i(X)$ of level $\leq i - 2j$.

Source. folder 12, pp. 2–3, 41–42, 52–54, inventory dating 1964–[c. 1971]; folder 36, pp. 12, 16. The pages do not name the conjecture; the name is the reading's.

The route. Page 3 states the expected equality and draws its consequence, granting resolution of singularities and the standard conjectures « (conséquence de la résolution et des conjectures standard) »: the dimensions of the $\mathrm{Filt}^j H^i$ are independent of ℓ, since traces of motivic endomorphisms are rational and independent of ℓ (pp. 29–30). One inclusion comes from the coniveau spectral sequence, read at E_1: for X smooth projective the classes supported in codimension p are Gysin images, of level $\leq n - 2p$ (p. 41). For the level itself, pp. 52–54 read weight and effectivity on the eigenvalues of Frobenius, and folder 36 applies this after reduction to a finite field: a piece of level 1 has eigenvalues divisible by q^{n-1}, so a p-adic unit eigenvalue shows that a piece is not of level 1 (p. 16).

Where it stops. The reverse inclusion, that the largest sub-motive of low level lies in the coniveau filtration, is never argued, and the Frobenius characterisation of p. 54 is struck. In folder 36 he cannot show, even granting the standard conjectures, that the level of a motive can only drop under specialisation: « la triste vérité est que je ne sais pas » (p. 16).

Today. Open. Grothendieck's 1969 paper corrected Hodge's general conjecture of 1950, which is false, into this form. In its Hodge-theoretic form, that the coniveau filtration equals the largest Hodge substructures of bounded level, it holds for all powers of the Jacobian of a very general complex curve of genus ≥ 2 [2, Prop. 6.5]. The OpenAI preprint on abelian varieties with complex multiplication deduces it for them, not beyond [40].

Conjecture 3 (The period conjecture)**.** Over $\overline{\mathbb{Q}}$, the functor that sends a motive to its de Rham realisation, its Betti realisation and the comparison isomorphism between them is fully faithful. In the form of Bost and Charles, the comparison isomorphism is Zariski-dense in the torsor of motivic periods, so that the periods of a motive generate a field of transcendence degree equal to the dimension of its motivic Galois group [6, Conj. 2.12]; with the standard conjectures, this form implies the first [6, Prop. 2.13].

Source. folder 13, *Groupe de Galois motivique*, pp. 30–34, 47–51, dated 1965; folder 18,

pp. 10–12, 41–50, inventory dating 1969–[c. 1972]. An undated plan headed « Motifs » (IHÉS, fonds Grothendieck–Illusie, 2ILL5, p. 8) states an arithmetic variant, marked « (de Groth.) », perhaps in another hand: for X of finite type over $\mathbb{Z}$, the functor that sends a motive to its de Rham realisation on $X_{\mathbb{Q}}$ and its crystalline realisations with Frobenius at every p, in place of the Betti realisation, is fully faithful.

The route. Folder 13 starts from a negative observation: the algebraic data of a realisation do not say which Hodge data come from motives, and must be completed by conditions of the type « arg. de transcendance » (p. 31). Pages 33–34 prove an abstract lemma: if g places the second lattice and H is the group generated by the closure of the elements ts^{-1}, then $V \cap T'(V) = V^H$, so full faithfulness becomes $V^H = V^G$ for every representation V. Page 47 asks the question without filtration or Hodge structure, reduces it at once to $\overline{\mathbb{Q}}$ and to V simple, and settles rank one: « c'est une puissance de Tate, et ça marche ». Folder 18 turns it into transcendence: the field generated by a generic period point has transcendence degree $\dim G - 1$ for a group G attached to the motive (p. 11), which « contient le th. de Schneider » for elliptic curves without complex multiplication; elliptic curves are the test case (p. 12), and pp. 46–50 sketch an adelic analogue.

Where it stops. No non-Tate motive is treated. For a product of two non-isogenous elliptic curves without complex multiplication his bound gives at most 5 against the expected 6 (folder 18, p. 44). Over a field of finite type the functor is not fully faithful, and the page breaks off on « idiot ! » before hoping for a weaker result (folder 13, p. 48).

Today. Open. For a Weierstrass function with algebraic invariants, two of $\omega_1, \omega_2, \eta_1, \eta_2$ are algebraically independent, and so are π/ω and η/ω for any period ω and its quasi-period η [11, §2 A, Th. 1 and 3]. For an elliptic curve with complex multiplication this proves the conjecture, the periods and quasi-periods having transcendence degree 2; Bost and Charles cite it as the main known case [6, §2.2.2]. Without complex multiplication, the expected degree 4 is open. The conjecture's consequence for classes, that a class of degree $2k$ rational in both the Betti and the de Rham structure (up to $(2\pi i)^k$) is algebraic, is proved in codimension 1 for products of curves, abelian varieties and K3 surfaces, and in codimension 2 for cubic fourfolds [6, Cor. 3.9, 3.10, Th. 5.6]. A Lean statement of the elliptic case is in the Tau Ceti roadmap on elliptic periods.

Conjecture 4 (The homotopy hypothesis)**.** *n-groupoids, up to n-equivalence, are the same as homotopy n-types, and ∞-groupoids the same as homotopy types.*

Source. folder 134-2, pp. 42–46, postscript of a letter to Breen of 5 February 1975; pp. 48–51, letter of 17 February; pp. 145–149 and 167–170, French original of the letter of July 1975; folder 134-1, p. 6, letter to Deligne of 27 August 1974.

The route. To a Kan complex K he attaches an n-category $C_n(K)$ of points, paths and homotopies, the top level taken up to homotopy, and notes that $\pi_i(K) \cong \pi_i(C_nK)$ for $i \leq n$ almost by definition (pp. 42–43); the two localised categories should then be equivalent. He uses the hypothesis to choose the definitions: the yoga « pourrait constituer un fil d'Ariane » for defining non-strict n-categories, their functors and their equivalences (p. 43). He checks the dictionary on familiar constructions (p. 45): loops and $\mathrm{Hom}(x, x)$, mapping spaces and functor categories, homotopy fibres and fibre products, $K(\pi, n)$ and n-gerbes, truncation and classes

of top arrows. The abelian case comes first: strict Picard n-categories should be complexes of length n, and the correspondence between simplicial sets and ∞-groupoids a set-theoretic Dold–Puppe (pp. 146–147); for groups in n-categories he wants a « non-commutative chain complex », and for $n = 1$ gives crossed modules with their invariant in H^3 (pp. 148–149). The functor from simplicial sets to n-groupoids is not negotiable: a theory without it « me semblerait canularesque » (p. 146). On the topos side, locally constant stacks on X should be functors on $\Pi_n(X)$, and Π_n recovered from them as in Galois theory (pp. 48–51, 167–170).

Where it stops. Nothing is proved: the definition of non-strict n-categories is the missing piece, and he writes of the « inquiétant mirage » of n-limits (p. 169, reading uncertain). He cannot define the classifying topos of an n-gerbe for $n \geq 2$ (p. 45), and asks Breen for candidates for the non-abelian complexes. *Pursuing Stacks* (1983), not transcribed in the edition, takes the programme up with coherators and test categories.

Today. The answer depends on the model. It is false for strict n-groupoids from $n = 3$: for any realisation functor that preserves homotopy groups, no strict 3-groupoid realises the 3-type of S^2, and no strict ∞-groupoid realises S^2 [47, Prop. 5.1, Cor. 6.1]. Simpson conjectured that strictly associative composition with weak units suffices [47, §7, Conj. 1]; Henry proves a form of this for ∞-groupoids in which only the « regular » compositions are strict [25, §§6.1–6.3]. For Grothendieck's globular ∞-groupoids in Maltsiniotis's formulation it is open. Henry reduced it, for a fixed coherator, to one statement: on a finitely cellular ∞-groupoid, freely adding a cell parallel to an n-cell and an $(n+1)$-cell between them is a weak equivalence [24, Conj. 5.3.3, Cor. 5.3.13]. An OpenAI preprint of 24 September 2026, not peer-reviewed, announces a proof for the Ara–Henry coherators, of which only the elementary-expansion step is formalised in Lean [40].

Conjecture 5 (The section conjecture)**.** For a hyperbolic curve U over a field K of finite type over $\mathbb{Q}$, every section of $\pi_1(U) \to \mathrm{Gal}(\bar{K}/K)$ comes, up to conjugacy, from a rational point of U or from a rational point at infinity.

Source. folder 149, pp. 26–127, undated: Theorem 3 (pp. 31–33), the « Conjecture fondamentale » (p. 47), Conjectures A, B and B′ (pp. 55–62), relative curves (pp. 63–64, where « Nos conjectures impliquent donc qu'il y a exactement deux sortes de telles sections »), specialisation (pp. 69–98), the birational form (pp. 123–127); folder 142, pp. 79–81.

The route. Only injectivity is proved, and arithmetically: he first tried a geometric proof, « Finalement, je n'arrive pas à le prouver par voie géométrique » (p. 26). He reduces to a semi-abelian variety and the Kummer sequences for all n; a point with trivial class is infinitely divisible, hence zero since the group of rational points is finitely generated (pp. 31–33), and one prime ℓ is not enough (p. 33). The section conjecture is the case « V a point » of his conjecture on morphisms (p. 48). The rest of the folder reformulates the conjecture. Sections at a cusp are told from rational points by filling the hole (p. 54). Conjecture B recasts the statement as a fixed point: the action of the Galois group on the universal cover completed by its cusps has « un pt fixe et un seul » (p. 58). Over a model the sections become lifts to $\pi_1(S)$, and a local criterion says that a rational point extends over a discrete valuation ring exactly when its section kills the inertia (p. 73); this compares sections over K and over the residue fields. Pages 98–103 try to build counterexamples from bad reduction, without success.

Where it stops. No route to surjectivity appears. Mordell's conjecture, which Faltings proved in 1983, is never used; the only arithmetic input is the Mordell–Weil theorem, for injectivity. The case of the thrice-punctured sphere at a cusp rests on a relation the page calls « hautement plausible » (p. 60). On reduction he writes that he has no idea for the moment (p. 102), and the explicit cocycle of a point is called fascinating work not yet begun (folder 142, p. 81).

Today. Open. Injectivity is known, as are a birational p-adic form (Koenigsmann) and the real analogue. An OpenAI preprint (24 September and 6 October 2026), not peer-reviewed and not formalised, announces the local p-adic conjecture for curves of genus at least 2, and the global one over number fields when a descent locus equals the rational points, which covers $X_0(N)$ and $X_1(N)$ of genus at least 2 over $\mathbb{Q}$; the affine, cuspidal form of folder 149 is not claimed [40].

Conjecture 6 (Ihara's question: $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and the Teichmüller tower)**.** The outer action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the profinite fundamental groups of the moduli spaces $M_{g,\nu}$ identifies it with the automorphisms of the tower that respect its structure; in Drinfeld's form, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \widehat{GT}$ is an isomorphism.

Source. folder 142, pp. 12–16, 21–41, 86–100, inventory dating [from 1978]; folder 143, pp. 28–30; folder 145, pp. 4–37; folder 147, pp. 59, 103, 148–151; folder 144, p. 4; folder 146; folder 67, p. 49, January 1984.

The route. The target is stated in folder 143: « On aimerait » that $\pi_1(M_{g,\nu})$ map isomorphically onto the outer automorphisms of the profinite surface group that satisfy the loop condition, which would describe the Galois group through them (p. 29). Folder 142 reduces the action on any curve to the thrice-punctured sphere, by Belyi and a Lefschetz-type curve (pp. 86–88), and lists the necessary conditions for an automorphism of the profinite groupoid to come from Galois: it lifts to the covers, stabilises the kernels of morphisms, is compatible with morphisms between covers, and satisfies the loop condition, « sûrement la plus importante de toutes, et de très loin ! » (p. 99). He conjectures them sufficient for one curve (pp. 94–95, Problem 12). Since the first conditions cannot be written down directly in the parameters (p. 98), he turns to the moduli spaces, with a conjecture tied to $M_{0,4}$ and $M_{0,5}$ that the page announces and that is nearly illegible (pp. 98–99). The low levels are computed: automorphisms commuting with $\mathfrak{S}_3$ as triples satisfying one relation (folder 142), pairs (p, g) with a marginal « $= \chi(u)$ » (folder 145), presentations in modular dimension 0 and 1 (folder 146). Folder 147 then plans the discrete tower: a more or less complete understanding of the cases of modular dimension at most 2 « donnera la clef pour une compréhension complète dans le cas général » (p. 59), a programme to present all the groups and their maps by generators and relations (p. 103), and a gluing theorem at infinity over the strata of codimension 1 and 2 (pp. 149–151, Problem 13).

Where it stops. The sufficiency of the conditions, the $M_{0,4}$–$M_{0,5}$ conjecture, the programme of folder 147 and the gluing theorem are announced, not proved. Page 59 of folder 147 speaks of the discrete groups, in the analytic setting; reading it as the two-level principle for the Galois action is the edition's. The last step, that an automorphism satisfying the relations of the tower comes from Galois, is not attempted anywhere in the notes.

Today. Open; the map is injective by Belyi's theorem. Hatcher, Lochak and Schneps add to

the relations of $\widehat{GT}$ with $\lambda = 1$ one relation in the profinite mapping class group of genus 1 with two boundary components, and show that the subgroup Λ it defines acts on every $\widehat{\Gamma}^m_{g,n}$, compatibly with the homomorphisms induced by the inclusions of subsurfaces, and injects into the special outer automorphisms of that tower [23, Th. A–C]. The relations defining Λ live in modular dimension 1 and 2, so relations on the first two levels already give an action on the whole tower, in all genera. On the motivic side, Brown proved that the motivic Galois group of mixed Tate motives over $\mathbb{Z}$ acts faithfully on the motivic fundamental group of $\mathbb{P}^1 \smallsetminus \{0, 1, \infty\}$, and notes that this implies the Deligne–Ihara conjecture on the outer action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on its pro-ℓ fundamental group [9, §1, Conj. 1, Th. 1.1]. An OpenAI preprint of 23 September 2026, not peer-reviewed, announces the Deligne–Drinfeld conjecture, that the rational Grothendieck–Teichmüller Lie algebra is free on one generator in each odd weight ≥ 3; it does not touch Ihara's question [40].

Conjecture 7 (The p-curvature conjecture)**.** Let S be integral, of finite type and dominant over $\operatorname{Spec}\mathbb{Z}$, X/S smooth, and E a vector bundle on X with an integrable connection. If the p-curvature of E vanishes at almost all closed points of S, then E has finite monodromy at the generic point.

Source. folder 93, pp. 188–212, *Connexions de Gauss–Manin et équations de Picard–Fuchs*, inventory dating [1972]–1981. No other folder has it.

The route. Page 194 notes, without proof, that finite monodromy forces vanishing p-curvature at almost all closed points, and asks: « La réciproque est-elle vraie ? » It then splits rank one: « La question précédente se décompose en deux ». First, is the class of E in the Picard group torsion? If so, $E = \mathcal{O}_X$ with connection $d + \omega$, and the question becomes whether a closed form that is locally logarithmic at almost all primes has a multiple that is logarithmic. In characteristic 0 the target is explicit: the logarithmic forms are a group $\mathbb{Z}^r$ by Rosenlicht, so the integrable classes form $(\mathbb{Q}/\mathbb{Z})^r$ (pp. 191–192). Pages 195–197 extend the question to torsors under group schemes, and for an abelian scheme rewrite it on the universal vector extension of the dual: a section that is divisible by p at every closed point, is it of finite order? He reduces it to a local-global statement about connected subgroups, and the case of unipotent rank 0 is the one on which, « si ce résultat était acquis », everything reduces to the semi-abelian case. Pages 198–212 compute the condition $\tilde{C}\omega = \tilde{\omega}$ on the Cartier operator and the Verschiebung of the Picard functor with connection; that these serve the question is our reading, not the page's.

Where it stops. The converse and each of its reductions stay questions. There is no Gauss–Manin case, no Chebotarev argument and no reduction to curves in the notes.

Today. Open. Katz proved it for Gauss–Manin connections (1972), and it holds for connections with solvable monodromy, rank one included (Chudnovsky for curves, André, Bost).

9 How the list was made, and its limits

The fourteen problems were chosen from the register of findings among the rows still open and outside what Lean reached, and with a statement precise enough to be proved or refuted. Rows whose only interest is a slip on the page, a dating, or a statement now known to be in

print were left out, as were the conjectures already in the published record, which §8 treats from the side of the routes the notes take, and the companion paper from the side of their first appearance [26].

Three limits apply to every entry. The transcriptions are first passes by a language model, and doubtful readings are marked but not all checked against the facsimile; a statement may rest on a misread word, and the entries say where we know this. The literature searches were made by the same means, with full texts where they were open and abstracts otherwise; an entry that says « not found » reports a search, not an absence. And the steps marked « ours » have not been checked by a specialist.

Corrections, references and proofs are welcome at the address above. The register of findings, the Lean files and the notices of the folders are on the project's site, `https://grothendieck.commutator.io/`.

Changes

26 October 2026: the pseudoline dichotomy of folder 80 proved (§7); the count of Kostant quadruples derived, and Problem 7 replaced by the question Grothendieck asked Kostant; Problem 4 narrowed after Breen's 1975 theorem; references corrected after a citation check. 29 October 2026: §8 added, on the routes the notes take to seven published conjectures; the arguments for Problems 5 and 6 written out step by step, with what a specialist should check; Problem 5 narrowed to the condition $A^0(Y_T)$, after a Lean proof of the two cases of the splitting; the OpenAI preprints on four of the seven conjectures noted. 31 October 2026: Problem 14 answered for every bigon-free arrangement, three pencils included, by a sign argument on chirotopes and the Folkman–Lawrence theorem; the sign argument and its rank-2 step proved in Lean, Folkman–Lawrence left outside; the sources of §8 completed from the texts read (issue #102). 1 November 2026: Problem 10 updated after Grothendieck's letter to Breen of 5 February 1975 (IHÉS, 1ILL21), which withdraws the « universal » argument of 1974, and what would settle it restated; the letter to Giraud of 1969 added to the source of Problem 4; the full-faithfulness conjecture of the IHÉS plan « Motifs » (2ILL5) added to the source of Conjecture 3.

Declaration

This note was drafted with a language model (Claude Opus 5.5, Anthropic) from the edition's registers: the findings, the modernised readings and the Lean files. The transcriptions and the modernised readings it relies on were also made with language models (Claude Opus 5 and Opus 5.5). The author received no funding for this work.

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